

FINAL SEMINAR · OCTOBER 2026

Multi-Page Visually Rich Document Understanding

A survey of system design and an empirical attribution of failure

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Roadmap

01 Background and motivation

02 Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding

Lewei Xu, Yihao Ding, Zihan Xu, Siwen Luo, Daochang Liu, Wei Liu

EMNLP 2026, main conference

03 Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution

Lewei Xu, Yihao Ding, Zihan Xu, Daniel Yitian Su, Daochang Liu, Siwen Luo, Yifan Peng, Wei Liu

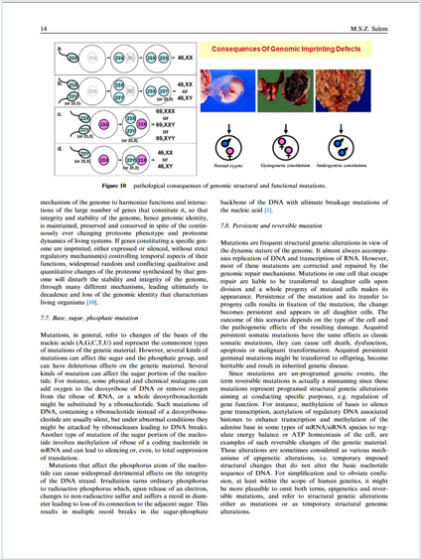
Submitted to EACL, under review

04 Proof of concept: a simple method for document understanding

Multi-Page Visually Rich Documents

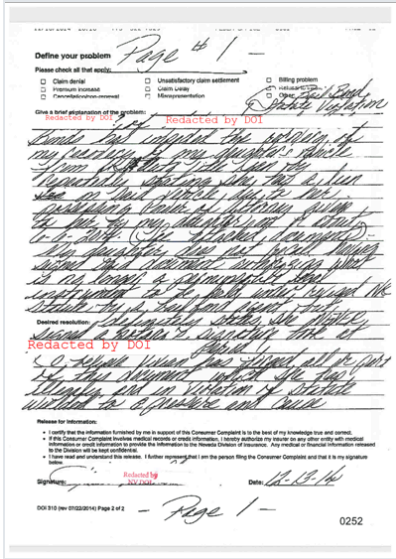
Meaning is carried jointly by **text**, **vision**, and **layout** rather than by text alone. **Multi-page** adds a further layer: meaning also runs between pages, as when a claim points to a figure many pages away.

Research paper
born-digital



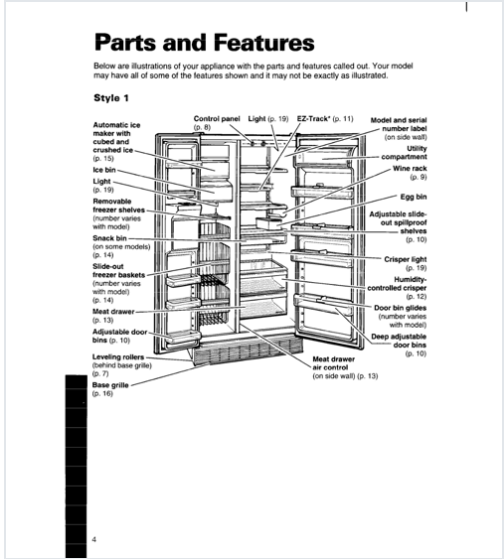
This research paper, titled 'Consequences Of Genomic Imprinting Defects', features a complex layout with multiple diagrams and text blocks. The diagrams illustrate the mechanisms of genomic imprinting and the effects of various imprinting disorders. The text is typeset in a standard serif font, with some sections highlighted in red. The paper is signed 'Page 1' at the bottom.

Signed form
scanned, handwritten



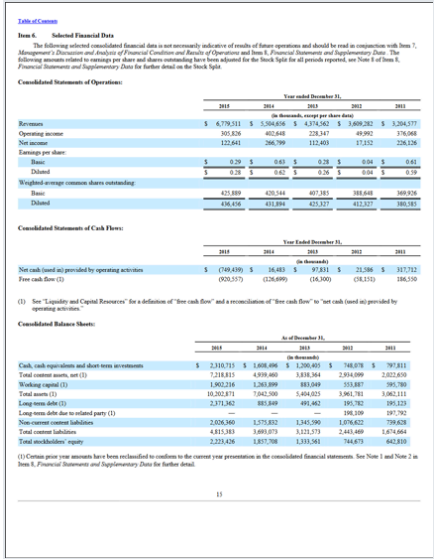
This is a scanned document, likely a form or a letter, with handwritten text in blue ink. The text is written on a pre-printed form with various fields and checkboxes. The document is signed 'Page 1' at the bottom.

Technical manual
layout-dependent



This is a technical manual for a refrigerator, titled 'Parts and Features'. It includes a detailed diagram of the refrigerator's interior and exterior, with labels for various components. The layout is highly structured, with a clear hierarchy of parts and features. The document is signed 'Page 1' at the bottom.

Financial report
dense tables



This is a financial report, titled 'Table of Contents'. It contains several tables of financial data, including 'Consolidated Statement of Operations', 'Consolidated Statement of Cash Flows', and 'Consolidated Balance Sheet'. The tables are dense with numbers and text, and are presented in a clear, professional layout. The document is signed 'Page 1' at the bottom.

Product brochure
photo-led



This is a product brochure for the Canyri SE Nightshade. It features a large, high-quality photograph of the car, with text describing its features and benefits. The layout is clean and modern, with a focus on the car's design and performance. The document is signed 'Page 1' at the bottom.

- **Text.** Small, typeset or handwritten, and legible only at far higher input resolution than a typical vision-language task requires.
- **Vision.** Charts, figures, photographs, and handwriting carry meaning that no text extraction recovers.
- **Layout.** Position is semantic, and reading order must be recovered rather than assumed.

What is Document Understanding?

Answer a natural-language question over a document, accurately.

topics, texts, and issues, building on others' ideas and expressing their own clearly and persuasively. 5. <i>Speaking and Listening 11/12.2:</i> Integrate multiple sources of information presented in diverse formats and media (e.g., visually, quantitatively, orally) in order to make informed decisions and solve problems, evaluating the credibility and accuracy of each source and noting any discrepancies among the data. 6. <i>Reading for Informational Text 11/12.3:</i> Analyze a complex set of ideas or sequence of events and explain how specific individuals, ideas, or events interact and develop over the course of the text. 7. <i>Reading for Informational Text 11/12.4:</i> Determine the meaning of words and phrases as they are used in a text, including figurative, connotative, and technical meanings; analyze how an author uses and refines the meaning of a key term or terms over the course of a text. 8. <i>Reading for Informational Text 11/12.7:</i> Integrate and evaluate multiple sources of information presented in different media or formats (e.g., visually, quantitatively) as well as in words in order to address a question or solve a problem. SUMMARY OF ALL KEY ASSIGNMENTS QUARTER 1: Unit 1 Predicting Your Success: TED TALK, GRIT: The Power of Passion and Perseverance; submit a written response on to Google Classroom. Unit 1 & 2 Team Activity Form five to seven teams. Each team is responsible for one of the sectors of the external business environment discussed in the chapter (economic, political/legal, demographic, social, competitive, global, and technological). (Interpersonal, Resources, Information) Quiz #1: Twenty core concepts from Unit 1 & 2 Critical Thinking Case Study--UBER Hits a Bumpy Road.: Unit 3 & 4 Team Activity Divide the class into teams. Each team should choose a country and research its infrastructure to determine how it will help or hinder trade. Include a variety of countries, ranging from the most highly developed to the least developed. Report your findings. -Resources, Interpersonal, Information, Technology) Critical Thinking Case Study--We Want Our MTV (International) Quiz #2: Twenty Core Concepts on Unit 3 & 4 ...Continued Unit 3 & 4 Team Activity Divide the class into teams. Each team should choose a country and research its infrastructure to determine how it will help or hinder trade. Include a variety of countries, ranging from the most highly developed to the least developed. Report your findings -Resources, Interpersonal, Information, Technology) Preparing for Tomorrow's Workplace Skills: - Do you have what it takes to be a successful franchisee? Working the Net: - Select three franchises that interest you. QUARTER 2: Preparing Working the Net: Business Plan - Visit Sample Business Plans at http://www.bplans.com to review examples of all types of business plans. - Select an idea for a company in a field that interests you and, using information from the site, prepare an outline for its business plan. Quiz #3- Comprehension of Twenty Core Concepts from Unit 5 & 6 Preparing for Tomorrow's Workplace Skills: Self Reflection (1 paragraph per question submitted into Google Classroom) Unit 7 Ethics Activity-Ethics Report: Training IT Replacement Working the Net Using a search engine, look for the term "company organizational charts," and find at least three examples of organizational charts for corporations, not-for-profits, or government agencies. Unit 8 Key Assignments: Quiz #4-Comprehension of Twenty Core Concepts from Unit 7 & 8 Critical Thinking Case Study: Discrimination in the Workplace Continues--submit on Google Docs Semester Final Assessments: Part 1: Ethics Self-Assessment Part 2: Comprehensive Final on Key Concepts from Unit 1-8 - Students will be in 8 teams of 4 assigned to 1 Unit each. - They will choose the key concepts from each unit to teach and add to the final. (A form of Reciprocal Teaching)	- Research them QUARTER 3: Preparing for Tomorrow's Workplace Skills-INTERVIEW - Are you motivated more by intrinsic rewards (satisfaction, sense of accomplishment, etc.) or by extrinsic rewards (money, bonuses, etc.)? Explain. - Interview some friends and classmates to find out what motivates them. Discuss your differences in perspective. (Interpersonal, Information) (Submit to Google Classroom through the provided Graphic Organizer. Team Activity - Assemble a team of three to five students. Imagine that you are the management team for a start-up business with limited resources but a need for a highly motivated, skilled workforce. Quiz #5: Comprehension of Twenty Core Concepts from Unit 9 & 10 Preparing for Tomorrow's Workplace Skills Use the Google search engine, http://www.google.com, to conduct a search for "supplier information," and visit the websites of several firms (for example, Walmart, Northrop Grumman, Verizon, etc.). Compare the requirements companies set for their suppliers. How do they differ? How are they similar? Preparing for Tomorrow's Workplace Skills Pick a small business in your community. Preparing for Tomorrow's Workplace Skills Pick a specific product that you use frequently, such as a cosmetic or toiletry item, snack food, article of clothing, book, computer program, or video game. Using the Net--Considering a career in marketing? Quiz #6- Comprehension of Twenty Core Concepts from Unit 11 & 12-Think of a product that you use regularly. Working the Net--Visit Industry Week's website at http://www.industryweek.com. Under Archives, do a search using the search term "supply-chain management." QUARTER 4: Preparing for Tomorrow's Workplace Skills Team Activity: Should companies outsource IT? Some executives believe that IT is too important to outsource and that application service providers don't have a future. Yet spending for ASP subscriptions, MSPs, and other forms of IT outsourcing such as cloud computing continue to grow. Working the Net--Learn more about the CERT Coordination Center (CERT/CC), which serves as a center of internet security expertise. Unit 15 & Unit 16 Key Assignment: -Build Key Portfolio Stock Project: your Investment Firm has just received \$100,000 to invest in 5 stocks FINAL (Introduced at the Start of 3rd QTR): Your Career in Business PART 1: PERSONAL REFLECTION: - How can you enhance your interpersonal skills? - Why is learning to plan so important in school and in the real world? - What skills should you develop in school that can transfer easily to your professional life and make it a success? - What are some strategies that will help you find, keep, and advance in your dream job? - What key attributes do employers look for when interviewing job candidates? PART 2: Student Companies FINAL ASSESSMENT: WORK-BASED LEARNING INTERNSHIP W/ AMAZON & OTHER PARTNERED LOGISTICS CENTERS
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Writer-ER Kackery/Cajon/SBCUSD

Writer-ER Kackery/Cajon/SBCUSD

“How many quizzes are there in the entire course?”

Answer: 6

- Find the two pages out of seventeen.
- Read a table that runs across the page break.
- Combine: four on one page, two on the next.

A course syllabus, pages 15 and 16 of 17 · a real question from MMLongBench-Doc

LLM Assistants

Provide a demonstration of how you would use tool calls to answer a difficult question over this long visually rich document.

Designing a sample tool-based walkthrough for a cross-page question.

I'll demonstrate on a question that needs a figure and a table: "How often does the final pipeline stop at its earliest allowed point, and how does its page usage compare with the number of gold evidence pages?" First I'll map printed page numbers to PDF pages, then read the relevant pages as images.

Read a file, updated tasks ▾

Added task **Locate pages for Figure 4.2 and Table 4.2** ▾

Request

Description

Map printed page numbers to PDF pages using the contents list and the offset seen on page 27

Added task **Read figure and table pages visually** ▾

Request

Description

Read PDF pages 60-61 and extract the bar heights and token statistics

Added task **Verify and answer** ▾

Request

Description

Cross-check figure values against surrounding text, then answer

Completed task **Locate pages for Figure 4.2 and Table 4.2**

Started task **Read figure and table pages visually**

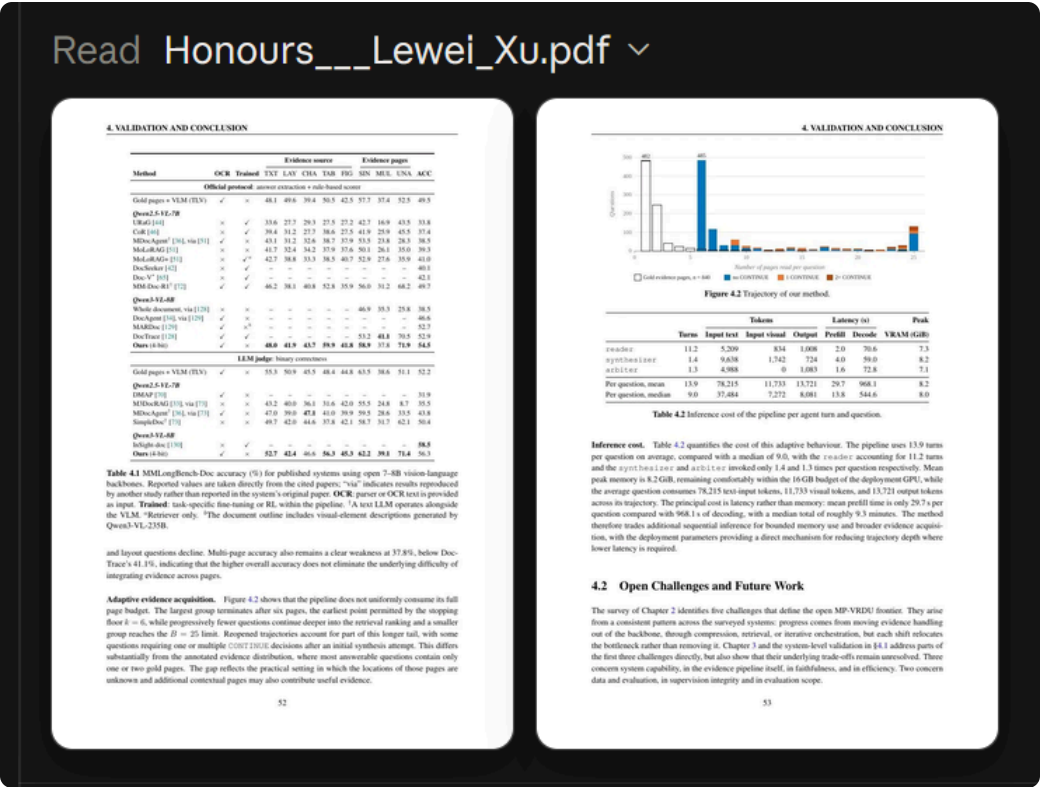
Read **Honours__Lewei_Xu.pdf** ▸

Verifying the data points plotted across the chart.

Completed task **Read figure and table pages visually**

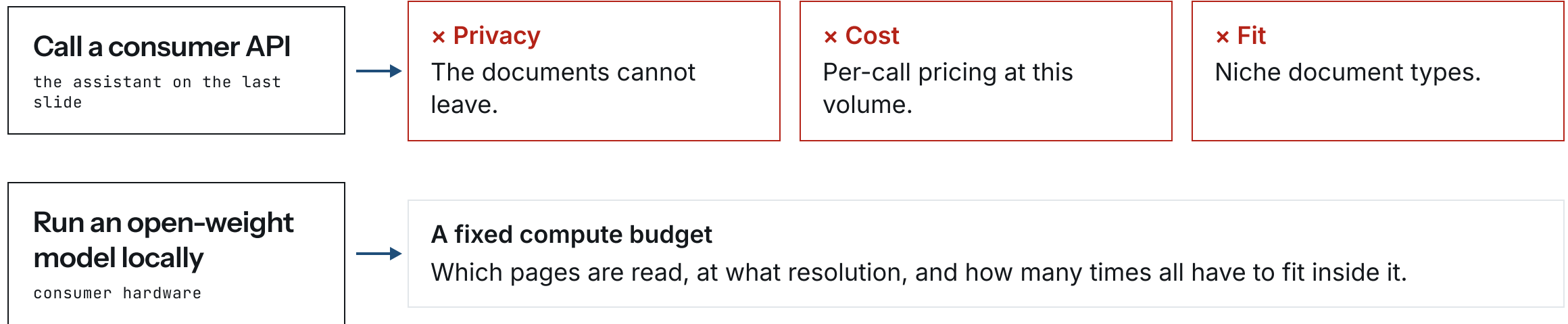
Completed task **Verify and answer**

- A question over a **100-page thesis**: the assistant plans, locates the pages, then answers.
- It **renders the pages and reads them as images**.



Real-World Constraints

A hospital, a mining company, a law firm: **tens of thousands of documents** to classify, summarise, or extract from.

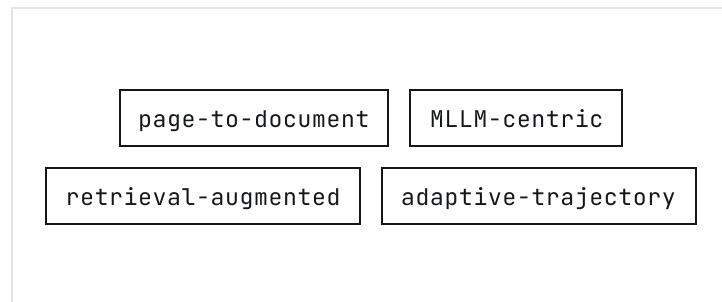


The research question: document understanding **under a fixed compute budget**.

Research Gaps

GAP 1

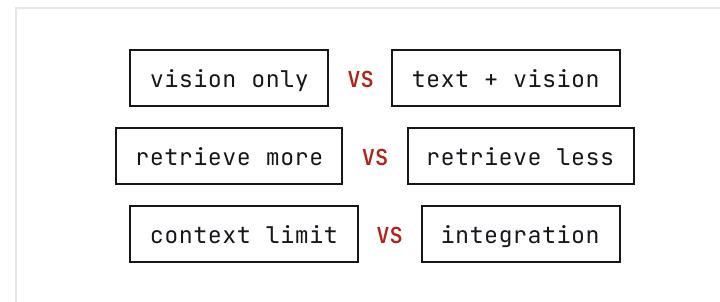
No unified view



Multi-page systems grew along **parallel lines**. No survey treats multi-page as its own problem.

GAP 2

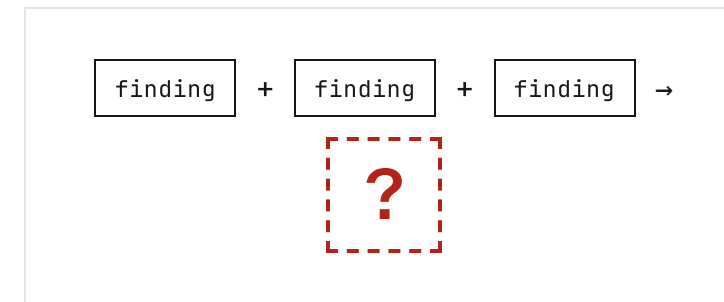
Competing, untested claims



Papers disagree on what matters, each on a **different pipeline and dataset**.

GAP 3

Advice not tested together



Design guidance is rarely validated in **one working system**.

Contributions

A thesis by compilation, in three parts. Each part answers one gap.

CONTRIBUTION 1

A survey

A taxonomy of the field through **evidence management**: four architectural families, and the strategies built around them.

EMNLP 2026, main conference

CONTRIBUTION 2

An empirical attribution

A framework of **three failure loci**, each tested by changing one thing in one pipeline.

Submitted to EACL, under review

CONTRIBUTION 3

A proof of concept

A training-free method built from the findings: **56.5%** on MMLongBench-Doc, the highest on its backbone.

Part 1: Survey of MP-VRDU

Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding

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Abstract

Documents spanning tens to hundreds of pages are pervasive in real-world workflows, from scientific papers to financial reports and technical manuals. In this setting, answer-relevant evidence is sparse and distributed across pages, and the document itself often exceeds a backbone MLLM's context window, making MP-VRDU a distinct problem from single-page VRDU. We survey MP-VRDU through the lens of evidence management, the organising principle that distinguishes multi-page from the single-page setting: the problem of deciding how document-scale evidence is selected and reasoned over when it cannot all be held at once. We introduce a taxonomy spanning four architectural families, multimodal representa-

MP-VRDU has seen rapid growth in recent years, with new architectures, training strategies, and inference methods emerging. Across these lines of work, the reasoning engine is almost always a general-domain MLLM; what varies is the evidence-management machinery built around it to handle documents whose length and structure exceed what the MLLM can process directly. Existing work has explored joint multi-page encoding (Tito et al., 2023), adaptation of pretrained MLLMs (Duan et al., 2025b), retrieval-augmented pipelines that decouple evidence selection from generation (Cho et al., 2024), and adaptive controllers that gather evidence iteratively (Sun et al., 2025). However, this body of work has developed along largely parallel lines, with limited cross-

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THE GAP

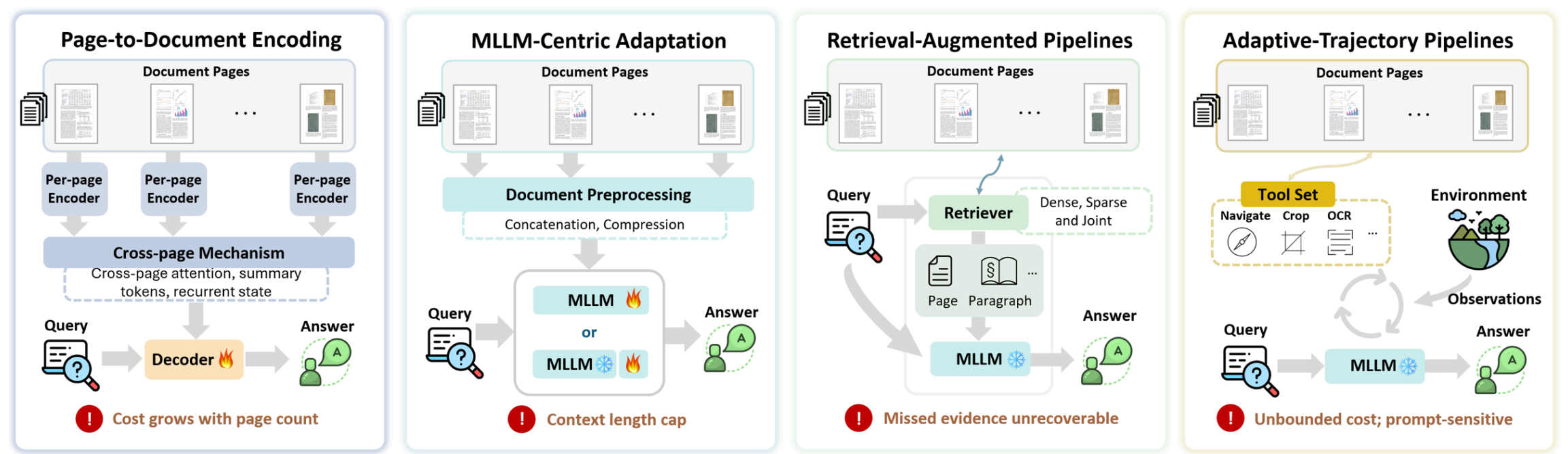
Work developed along **parallel lines**, with inconsistent terminology and little comparison across approaches.

CONTRIBUTIONS

- 1 Defines multi-page understanding as its own problem: **evidence management**.
- 2 A taxonomy: four architectural families, representation, inference-time and training strategies.
- 3 Consolidates the datasets and five open challenges.

Multi-Page Architectural Overview

The survey organises systems by **evidence management**, into four architectural families. **Page-to-document encoding models** jointly encode pages; **MLLM-centric adaptation models** fit multi-page input into an MLLM’s context and token budget; **retrieval-augmented pipelines** retrieve evidence before generation; **adaptive-trajectory pipelines** inspect evidence iteratively under an LLM controller.



Snowflake indicates typically frozen, fire indicates typically trained

Inference-Time and Training Strategies

Three evidence-management decisions. Each can be **prompted** around a frozen **MLLM**, or **trained** into it.

Retrieval and navigation

TRAINING-FREE

Similarity-based or relation-aware retrieval

TRAINED

Modality-alignment, cross-page-fusion, logical-relevance

Reasoning strategies

TRAINING-FREE

Chain-of-Thought, Self-Reflection, Self-Consistency

TRAINED

Trace-imitation, then reward-driven (GRPO)

Agentic methods

TRAINING-FREE

ReAct and tool use, multi-agent orchestration

TRAINED

Trajectory-distillation, reinforcement-learned

Part 2: Empirical Attribution Study

Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution

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Abstract

Multi-page visually-rich document understanding (MP-VRDU) requires managing evidence that is sparse, spread across pages, and often exceeds a model's context window. Prior work has produced competing, largely untested claims about how these systems should be built. We attribute incorrect answers to three failure modes, representation, selection, and reasoning, and isolate each over a multi-page document understanding dataset by intervening on one while holding the others fixed. We find that vision is necessary but does not replace text extraction, that missing pages bound accuracy while distractors cost little, and that reasoners fail to integrate evidence across pages even when it is fully supplied. Prompting can shift

et al., 2025) and that combining text and vision outperforms either modality alone (Han et al., 2025; Suri et al., 2025). A second disagreement concerns **selection**. Fixed-depth retrieval is brittle because a question can require more evidence pages than a fixed depth returns, while too many introduce distractors that dilute attention (Yan et al., 2025; Li et al., 2025). At the same time, other work identifies retrieval coverage as the main factor of accuracy (Liu et al., 2025; Zhu et al., 2025a). A third disagreement concerns **reasoning**. Performance drops on long inputs are attributed to context-length limits (Wang et al., 2024b; Liu et al., 2024), or to failures of evidence integration that persist even when the context is short enough to fit (Zheng et al., 2026; Guo et al., 2026).

Submitted to EACL, under review

THE GAP

Competing claims about what matters, reported on **different pipelines, datasets and settings**, so they cannot be compared.

CONTRIBUTIONS

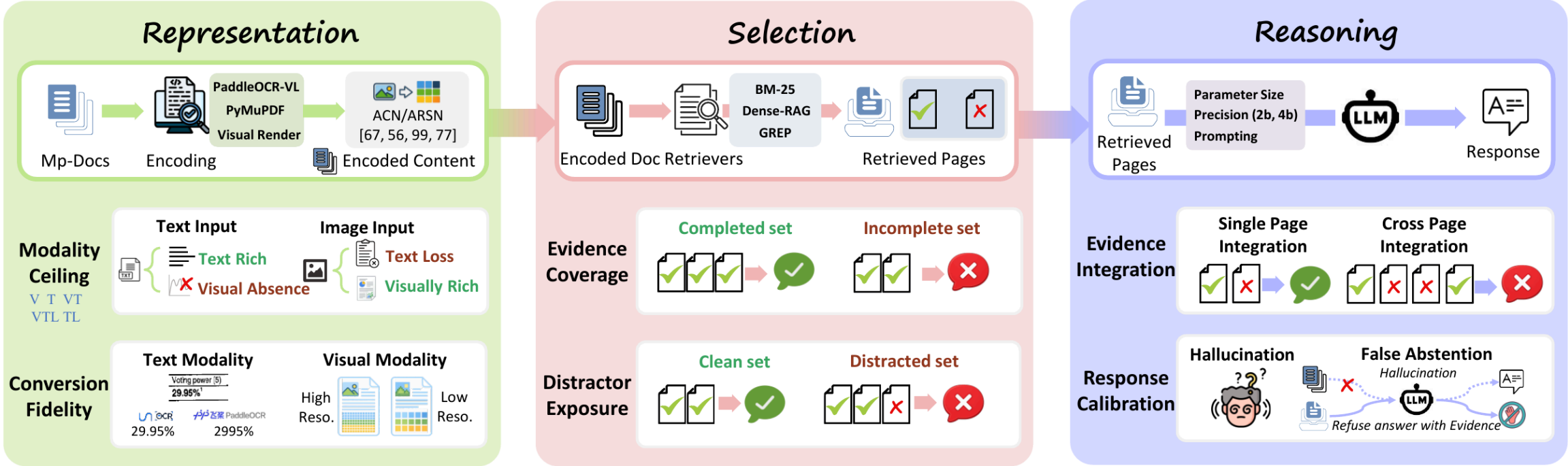
- 1 Three architecture-independent **failure loci**, with six mechanisms.
- 2 Controlled interventions in one pipeline: change one locus, hold the others fixed.
- 3 Transfer across datasets and models, with design implications and cost.

[illegible]

Correct answer: **6**. Suppose the system says **4**.

End-to-end accuracy **cannot tell these apart**, and each needs a different fix.

Attribution Framework



REPRESENTATION

Was the information preserved when the page was converted?

SELECTION

Did it actually reach the model?

REASONING

Did the model use it correctly?

Attribution by Construction

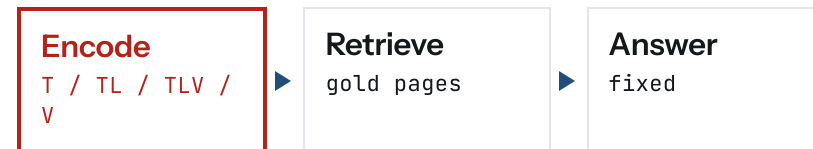
One simple pipeline. **Change one stage, hold the other two fixed.**



- **Gold pages** bypass retrieval, so selection can be taken out of the picture.
- **MMLongBench-Doc**: 135 documents, annotated with evidence pages, evidence type and answerability.

Modality Ceiling

Evidence	Abstention				Accuracy			
	T	TL	TLV	V	T	TL	TLV	V
Text	23.0	11.5	7.3	8.0	36.9	44.4	51.0	42.3
Layout	20.3	10.2	0.8	4.2	26.3	33.9	49.2	43.2
Table	15.9	7.2	6.7	9.1	35.6	44.7	44.7	34.1
Chart	32.6	28.1	7.3	6.2	16.3	16.9	42.7	40.4
Figure	36.6	24.8	7.6	7.9	14.5	22.1	42.4	39.0
<i>All</i>	26.3	17.2	6.2	7.1	29.0	35.7	49.5	42.6



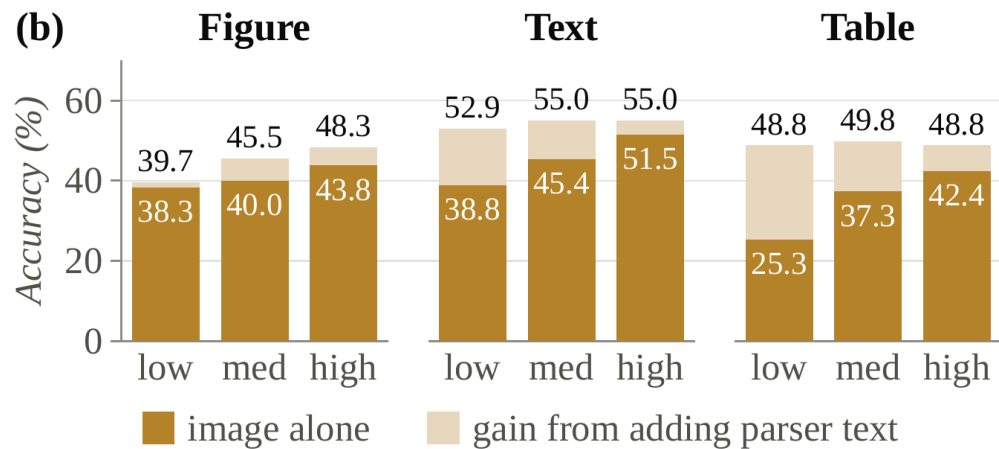
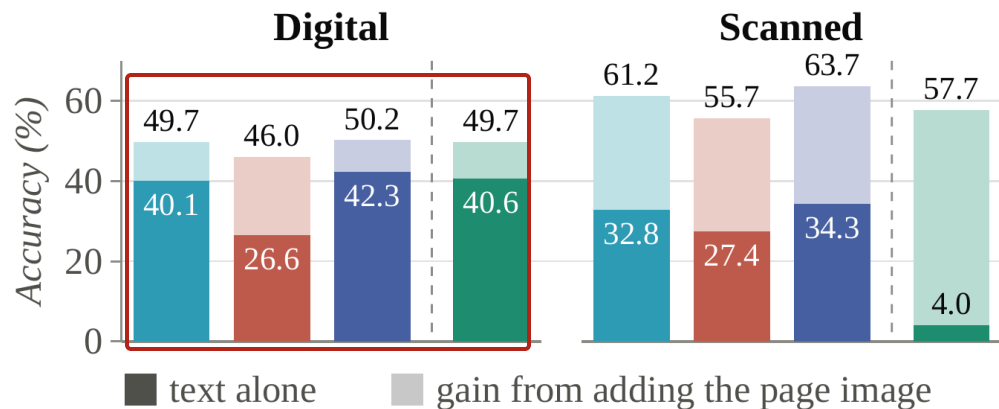
REPRESENTATION

Evidence accessibility depends on modality

- Text-only input refuses **chart and figure** questions most. Adding vision brings refusal **under 10%**.
- Images alone stay below text plus images: **42.6% against 49.5%**.

Conversion Fidelity

(a) PaddleOCR MinerU Unlimited-OCR PyMuPDF



Encode
parser, resolution

Retrieve
gold pages

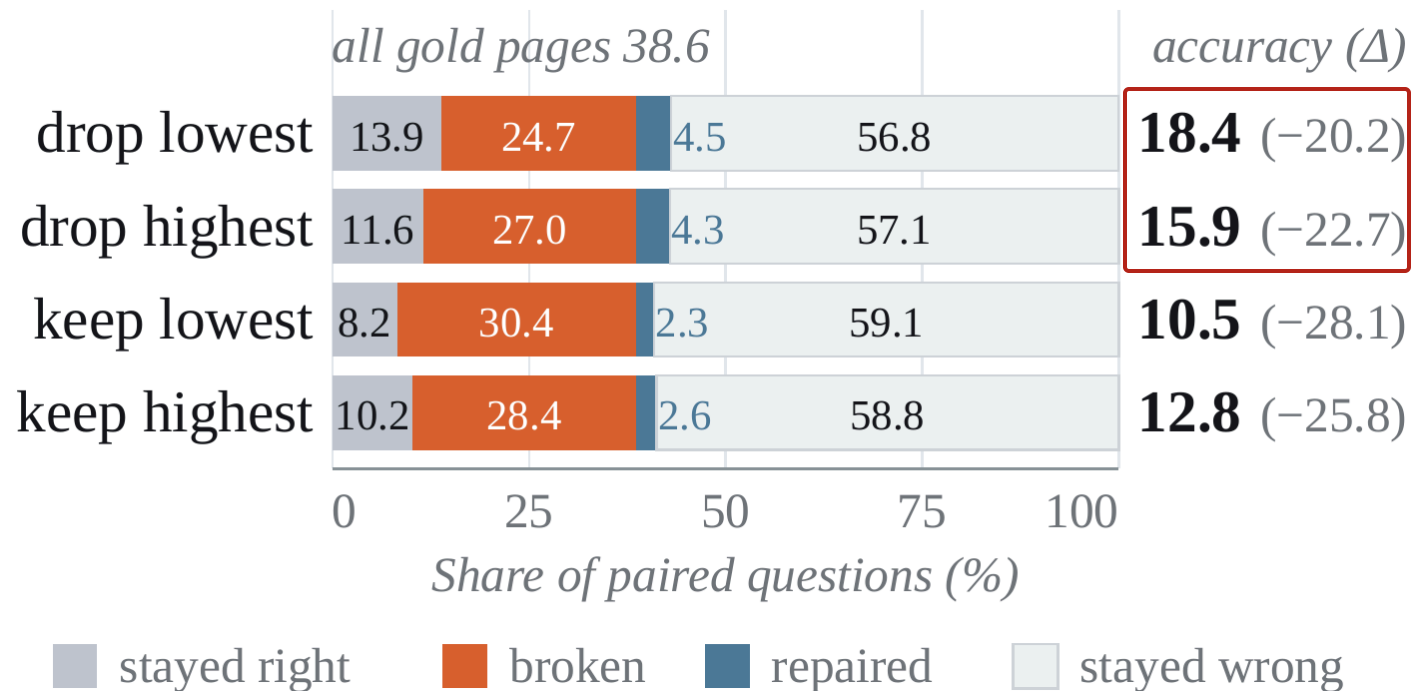
Answer
fixed

REPRESENTATION

Complementary modalities reduce conversion sensitivity

- The parser spread on digital documents shrinks from **15.7 to 4.2 points** once page images are added.
- Higher resolution helps images alone far more than text plus images.

Evidence Coverage

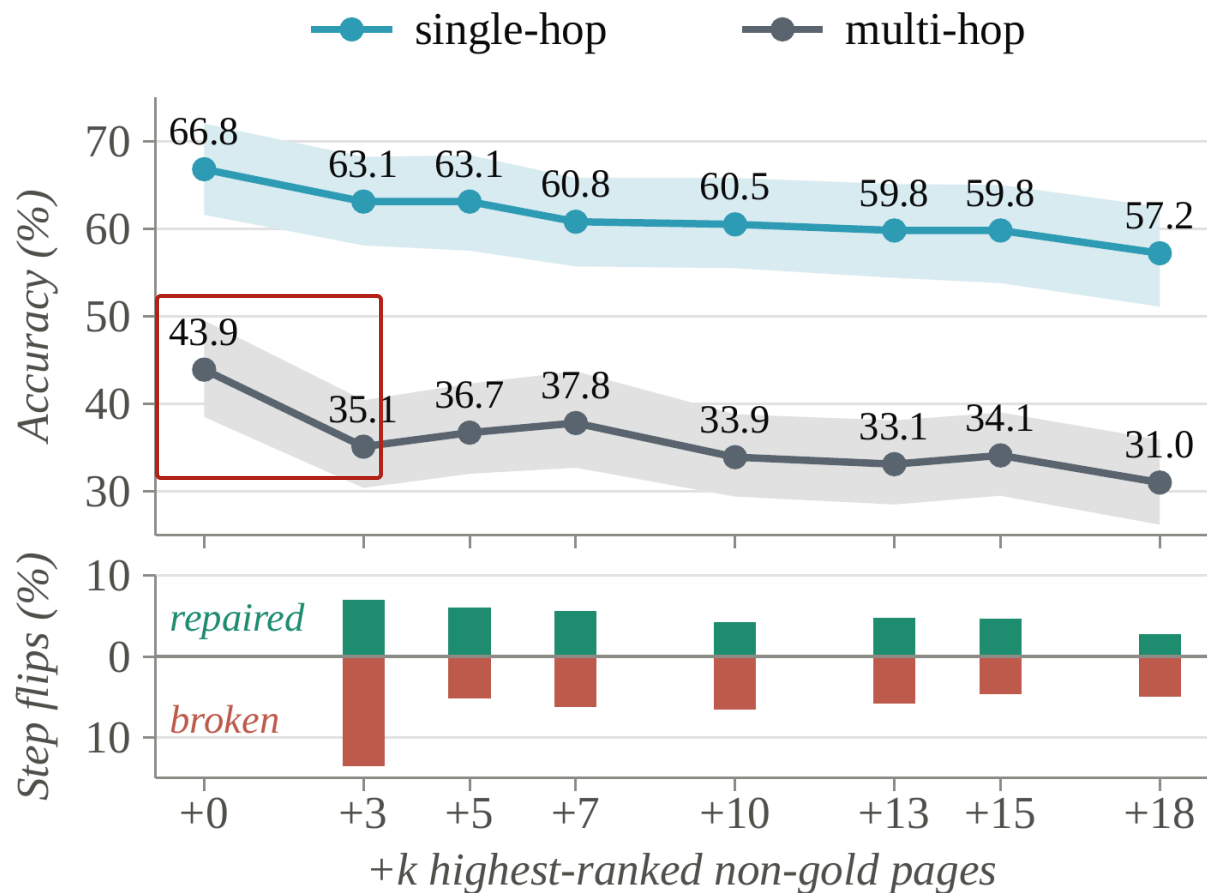


SELECTION

Omission causes large accuracy losses

- Removing **one gold page** takes accuracy from **38.6% to 18.4%** or 15.9%.
- Retrieval rank does not predict which page is safe to lose.

Distractor Exposure



Encode
fixed

Retrieve
gold + k other
pages

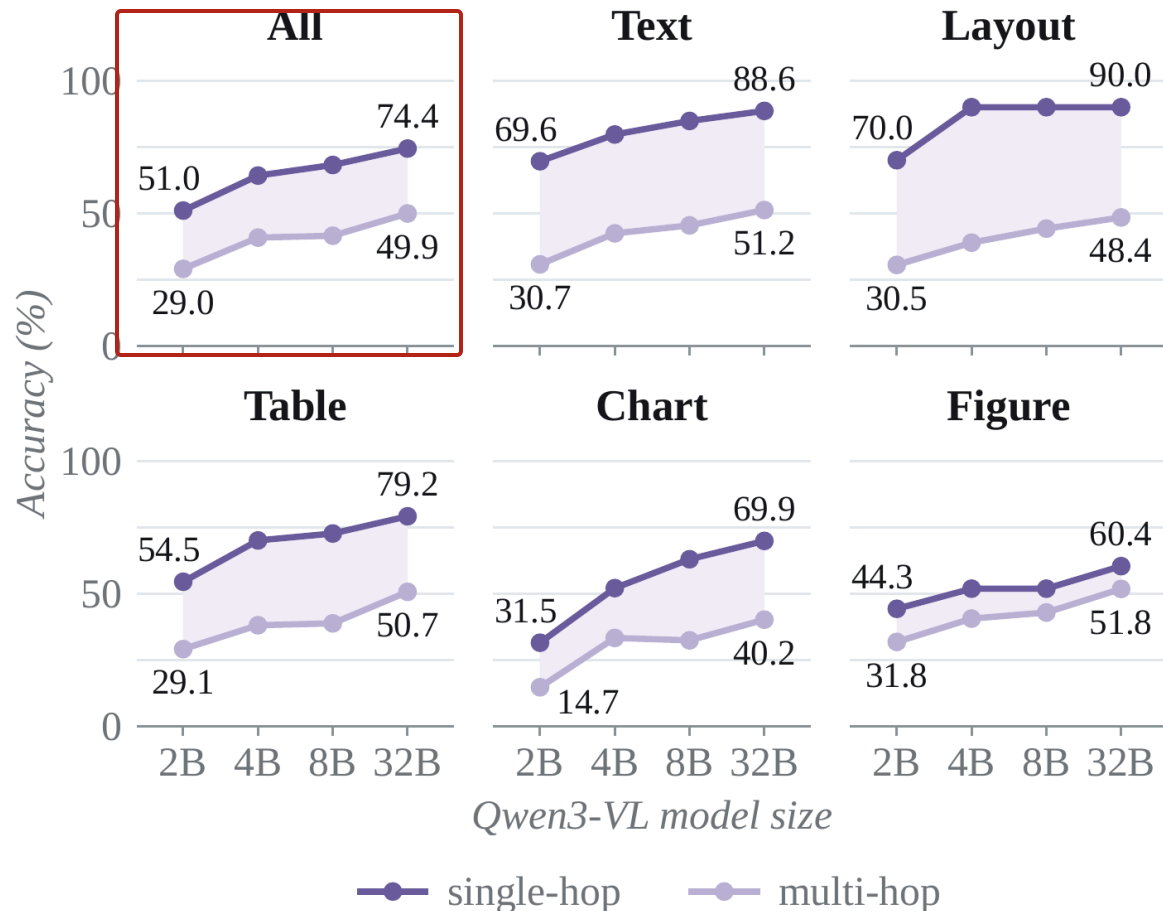
Answer
fixed

SELECTION

Multi-hop accuracy drops sharply with added context

- Three extra pages: multi-hop falls from **43.9%** to **35.1%**. Single-hop declines gradually.
- Every gold page is still there. Added pages both break and repair answers.

Evidence Integration



Encode
fixed

Retrieve
gold pages

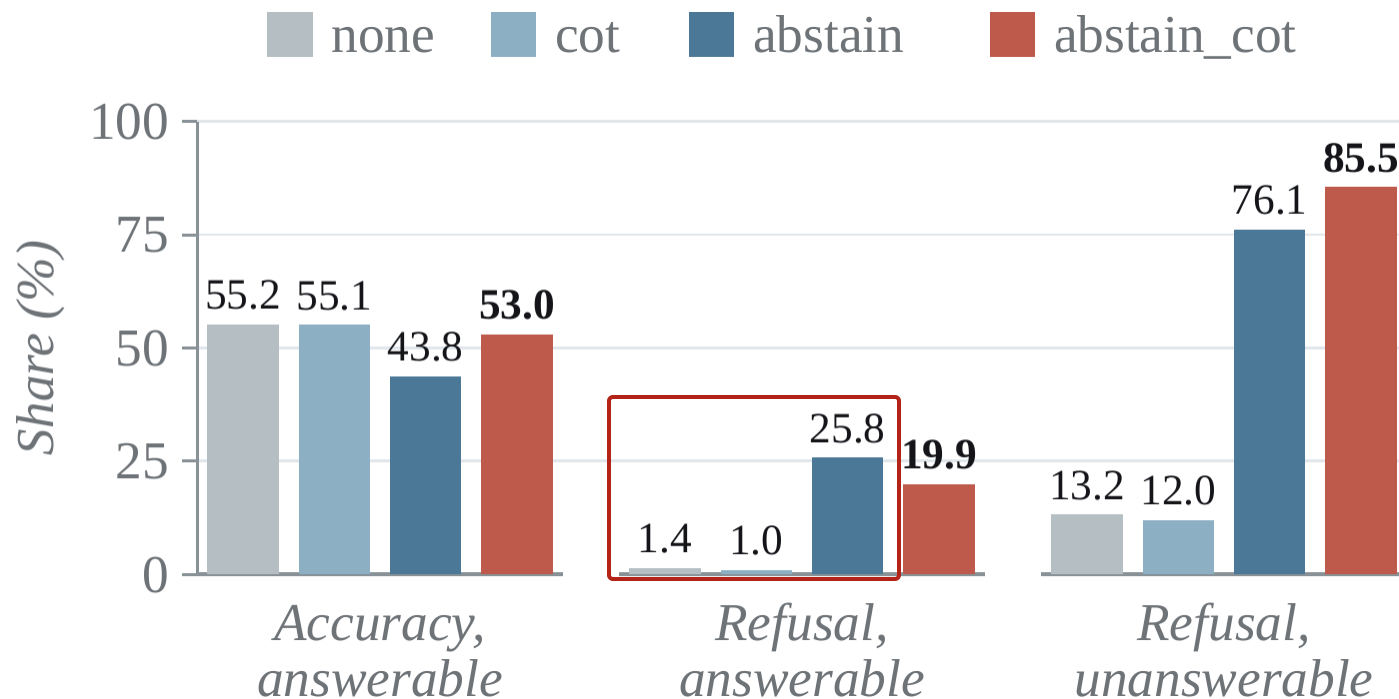
Answer
model size

REASONING

Scale improves multi-hop performance

- From 2B to 32B, multi-hop accuracy rises from **29.0%** to **49.9%**.
- Single-hop rises too, so scale helps evidence use in general.

Response Calibration

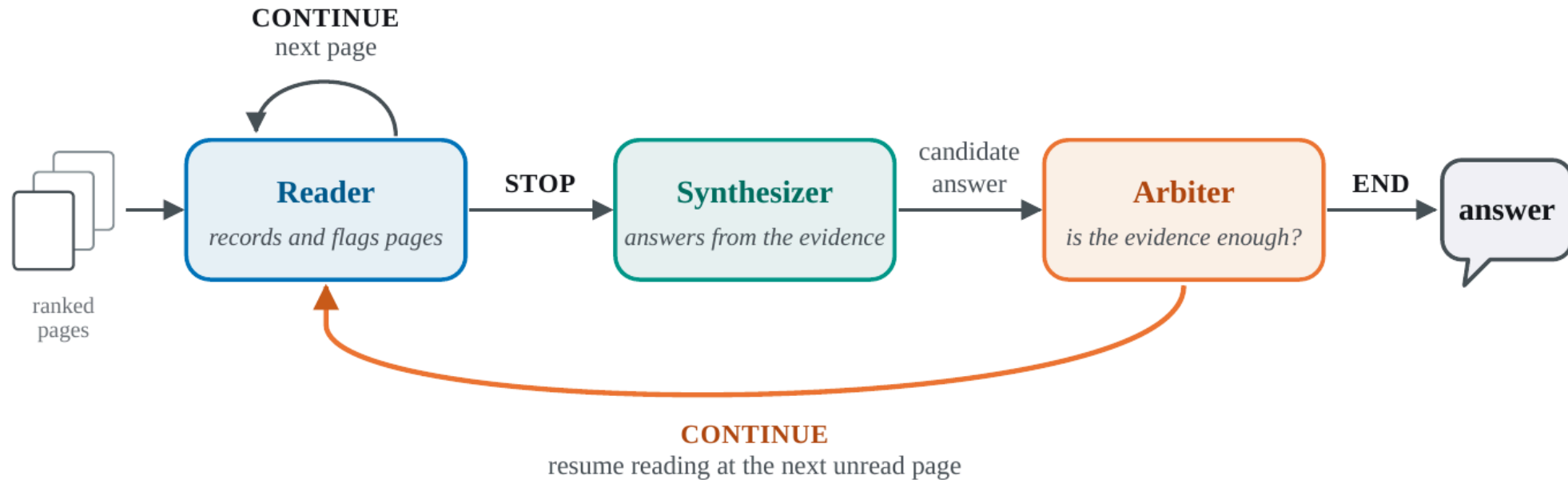


REASONING

Reasoning before abstaining limits false refusal

- An abstain instruction lifts correct refusal from 13.2% to 76.1%, but false refusal goes from **1.4%** to **25.8%**.
- Adding CoT recovers answerable accuracy to 53.0%.

Part 3: From Findings to a Simple Method



REPRESENTATION

Every page is read as **text and image together**.

SELECTION

Read broadly, then pass on compact records, not full pages.

REASONING

A separate arbiter decides **whether the evidence is enough**.

Simple Method Results

Method	OCR	Trained	MMLongBench-Doc				LongDocURL			
			SIN	MUL	UNA	ACC	UND	REA	LOC	ACC
Gold pages + VLM	✓	×	57.7	37.4	52.5	49.5	69.7	63.6	59.5	65.7
Whole document (via DocTrace)	×	×	46.9	35.3	25.8	38.5	–	–	–	45.1
M3DocRAG (via SLEUTH)	×	×	–	–	48.6	45.9 ^a	60.1	52.4	45.5	54.6
DocAgent (via MARDoc)	✓	×	–	–	–	46.6	–	–	–	–
MDocAgent (via SLEUTH)	✓	×	–	–	49.7	47.8 ^a	57.5	51.5	44.7	53.1
MoLoRAG (via SLEUTH)	×	×	–	–	51.7	48.8 ^a	63.4	52.7	49.3	57.6
SLEUTH	×	×	–	–	67.4	52.8 ^a	65.7	53.0	53.6	60.0
DocTrace	✓	✓	53.2	41.1	70.5	52.9	–	–	–	56.4
Ours (4-bit)	✓	×	60.5	39.2	76.9	56.5	67.2	66.3	47.9	61.3

- **56.5%** on MMLongBench-Doc and **61.3%** on LongDocURL: highest among the Qwen3-VL-8B systems, training-free, at 4-bit.
- Multi-page questions still trail DocTrace, **39.2** against **41.1**.

Conclusion

SURVEY

Organise

Multi-page document understanding is a problem of **evidence management**.

EMPIRICAL STUDY

Locate

Failures sit at **representation, selection, or reasoning**, and each needs a different fix.

PROOF OF CONCEPT

Validate

The findings hold up in a **working, training-free system**.

Thank you

Questions?

Retrieval and Navigation

Retrieval **narrows a long document to compact evidence units** (chunks, pages, or graph nodes) for downstream reasoning.

- **Similarity-based retrieval.** Scores each evidence unit independently against the query and returns the top-ranked subset. Sparse lexical signals such as BM25 are the baseline; dense retrieval uses vision-language late-interaction encoders such as ColPali and ColQwen, often with an LLM reranking stage after.
- **Relation-aware retrieval.** Retrieves along explicit document structure. Tree-based variants follow section hierarchies and outlines, so chunks keep their parent context; graph-based variants traverse typed edges between headings, entities, and cross-references.
- **The trade-off.** Independent scoring handles paraphrase and cross-modal queries, but cannot follow cross-page references or multi-hop dependencies. Relation-aware methods can, at the cost of parser dependence and traversal cost.

Evidence missed or poorly ranked at retrieval is **unrecoverable at generation**. What is retrieved still has to be reasoned over.

Reasoning Strategies

Reasoning strategies act over a **single evidence buffer assembled before the reasoning call**: the full document representation, a retrieved subset, or evidence from an agentic step. None updates model parameters.

- **Sequential reasoning: Chain-of-Thought.** Extends a single reasoning thread. CoT binds evidence from different pages into one chain and decomposes multi-hop questions into single-hop steps.
- **Sequential reasoning: Self-Reflection.** The model evaluates its own draft answer and revises it across rounds, optionally against a verifier or critic.
- **Parallel exploration.** Several reasoning paths over the same buffer. Self-Consistency samples multiple chains and takes the majority answer; sampling-adjudication replaces the vote with an LLM judge. Cost scales with the number of samples. Tree-of-Thoughts has seen little uptake.

A chain can only reason over the evidence already assembled. It **cannot recover what retrieval missed**.

Agentic Methods

Agentic methods put one or more **LLM controllers** in charge: either iterating evidence acquisition under inference-time control, or coordinating across specialised roles.

- **ReAct and tool-augmented reasoning.** A single controller interleaves thought, action, and observation. Actions are document-side tool calls (retrievers, parsers, OCR engines) or structured navigation over a pre-built document representation.
- **Multi-agent orchestration.** Modality-specialised decomposition gives text and visual evidence separate controllers plus a synthesis controller. Role-specialised decomposition splits planning, execution, verification, and synthesis.
- **What the adaptive trajectory buys.** Compute scales with question difficulty rather than document length, the single-step recall ceiling of retrieval-augmented pipelines is lifted, and the action sequence is an inspectable audit trail.

The cost is **controllability**: inference cost is hard to bound, and a controller can commit early to wrong evidence and continue down the wrong trajectory.

Training Strategies

Most systems adopt a backbone **already pretrained on single-page data**, so fine-tuning carries the weight. Training internalises the inference-time evidence-management decisions into model parameters.

Trained retrieval

Modality-alignment aligns visual page embeddings with OCR-derived text. **Cross-page-fusion** adds neighbouring-page context to each page representation. **Logical-relevance** ranks pages by their position in a cross-page reasoning path.

RETRIEVER

Trained reasoning

Trace-imitation supervises the backbone on demonstration chains in a fixed schema. **Reward-driven** training then refines it with GRPO, rewarding answer accuracy, evidence-page recall, and format compliance.

REASONER

Trained agentic methods

Trajectory-distillation clones a stronger teacher’s think-action-observation turns through SFT. **Reinforcement-learned** variants optimise the policy directly against trajectory-level rewards.

CONTROLLER

Two techniques recur across all three: **parameter-efficient tuning** (LoRA, adapters) and **multi-stage curricula** that go from single-page to document-scale supervision. All of it depends on the available datasets.

Datasets

Datasets serve three roles, and the multi-page setting makes all three harder: **per-page annotation does not scale** to documents of tens or hundreds of pages, and long PDFs are harder to source and license than single pages.

Pretraining

The exception rather than the norm. Confined to a few page-to-document and MLLM-centric systems that do continued pretraining on an existing backbone, because multi-page corpora are still small.

RARE

Fine-tuning

Has converged on LVLM-synthesised supervision over scraped PDFs, with reinforcement learning increasingly layered on top to shape evidence-grounding and trajectory behaviour.

LVLM-SYNTHESISED

Benchmarking

Three tiers: single-page extractive baselines, mid-length multi-page suites, and long-document multimodal benchmarks that need cross-page integration. The same small pool of PDFs gets reused.

CONTAMINATION RISK

Answer-only accuracy masks retrieval failure and rewards correct answers reached through incorrect reasoning. Newer benchmarks score retrieval and page-localisation as well, but it is not yet standard.

Experimental Setup

One simple pipeline throughout, with **one condition changed at a time** and everything else held fixed.

Encode

Every page becomes text, an image, or both.
Done once per document and cached, so no question changes what a page contains.

PaddleOCR-VL · 200 DPI

Retrieve

▶ A retriever scores every encoded page against the question and returns the top k , restored to document order.

ColQwen3 · page-level

Answer

▶ A multimodal model receives those pages and the question, and answers in a single pass.
Nothing is fed back.

Qwen3-VL-8B · greedy

- **The stages act once, and in order.** That is what lets a failure be attributed rather than guessed at.
- **Gold pages bypass retrieval.** For most tests the annotated evidence pages are supplied directly, so selection is neutralised and whatever remains is representation or reasoning.
- **Answers are scored by an LLM judge,** Gemini-2.5-Flash at temperature zero. It agrees with GPT-4o on 96.0% and with a human annotator on 98.3% of validation subsets.

Datasets and Evaluation

None of these experiments are possible without annotation that says **which pages support each question**. That is what picks the benchmarks.

MMLongBench-Doc

PRIMARY DIAGNOSTIC

135	47.5	847	244
documents	mean pages	answerable	unanswerable

The only multi-page benchmark we know of that annotates, per question, **the evidence pages, the kind of evidence it draws on, and the document's domain**, while also supplying plausible questions the document cannot answer.

Evidence pages make coverage measurable and define the oracle. Evidence source explains representation results by what the question actually needs. The unanswerable pool makes calibration testable rather than inferred.

Answers are scored by an LLM judge rather than exact match, since a correct response can differ in units, rounding, or formatting. The judge sees only the question, gold answer, and model answer, never the condition being tested. It agrees with GPT-4o on **96.0%** of a stratified sample and with a human annotator on **98.3%**. We track refusals and wrong answers separately, not just accuracy.

LongDocURL

SECONDARY REPLICATION

396	85.6	2,325	8
documents	mean pages	questions	unanswerable

Longer documents, used to check whether the findings hold under a **less demanding setting**. Born-digital throughout, almost no unanswerable questions, and many questions answerable from the text even when the annotated evidence is visual.

Its annotated pages are also not a minimal set: several may be redundant or independently sufficient, which we account for when reading its coverage and integration results.

LongDocURL Transfer

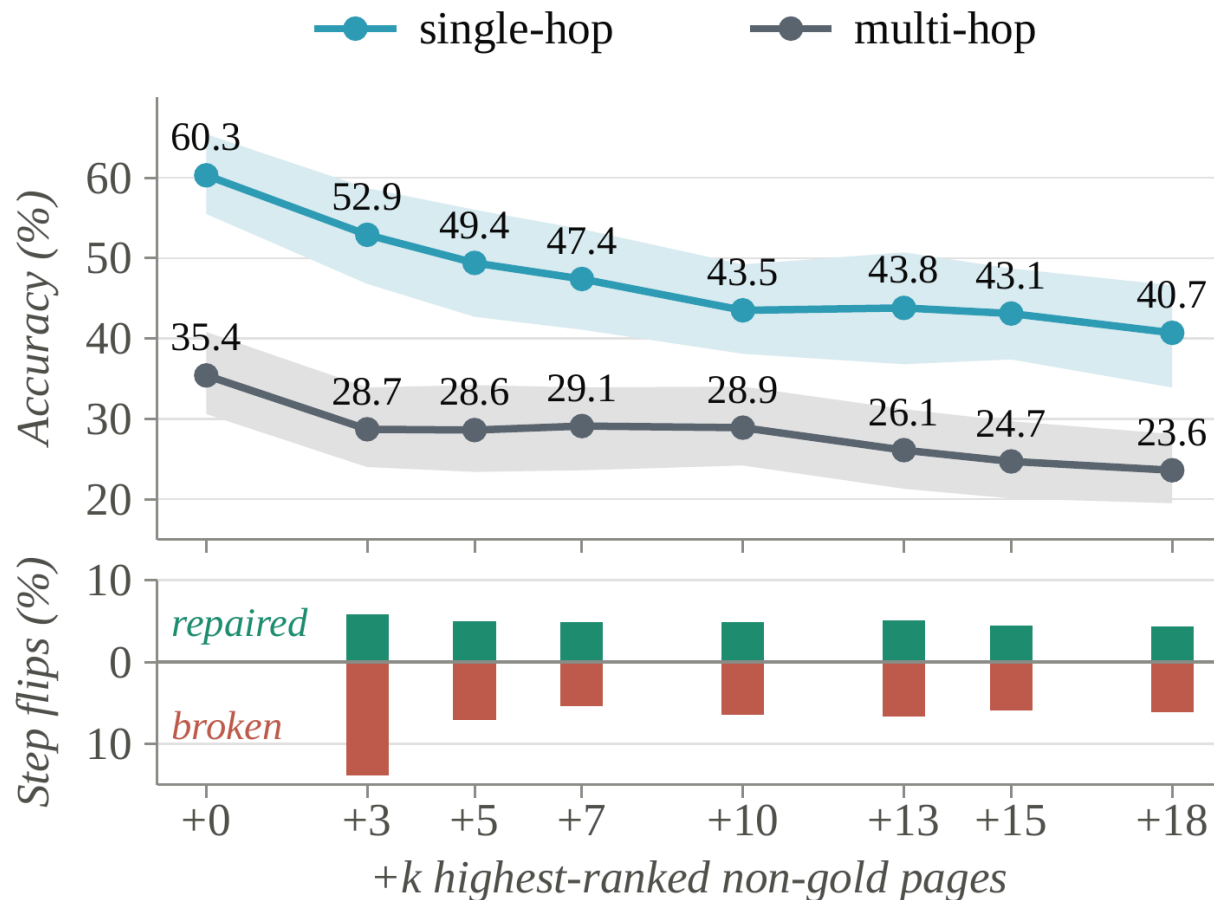
	T	TL	TLV	V
<i>LongDocURL</i>				
Text	75.3	79.5	82.3	74.0
Layout	48.7	52.4	56.3	52.2
Table	61.2	69.5	68.2	57.3
Figure	62.5	65.7	73.4	65.2
All	63.7	68.0	71.9	63.1
<i>MMLongBench-Doc</i>				
Born-digital	40.6	40.1	49.7	41.5
Scanned	4.0	32.8	61.2	58.2

REPRESENTATION

Born-digital documents leave less for vision to add

- Figure questions already reach **65.7%** from parser text. Adding images gives **3.9 points** overall.
- On MMLongBench-Doc, vision adds 28.4 points on scans and 9.6 on born-digital documents.

Distractors on Another Backbone

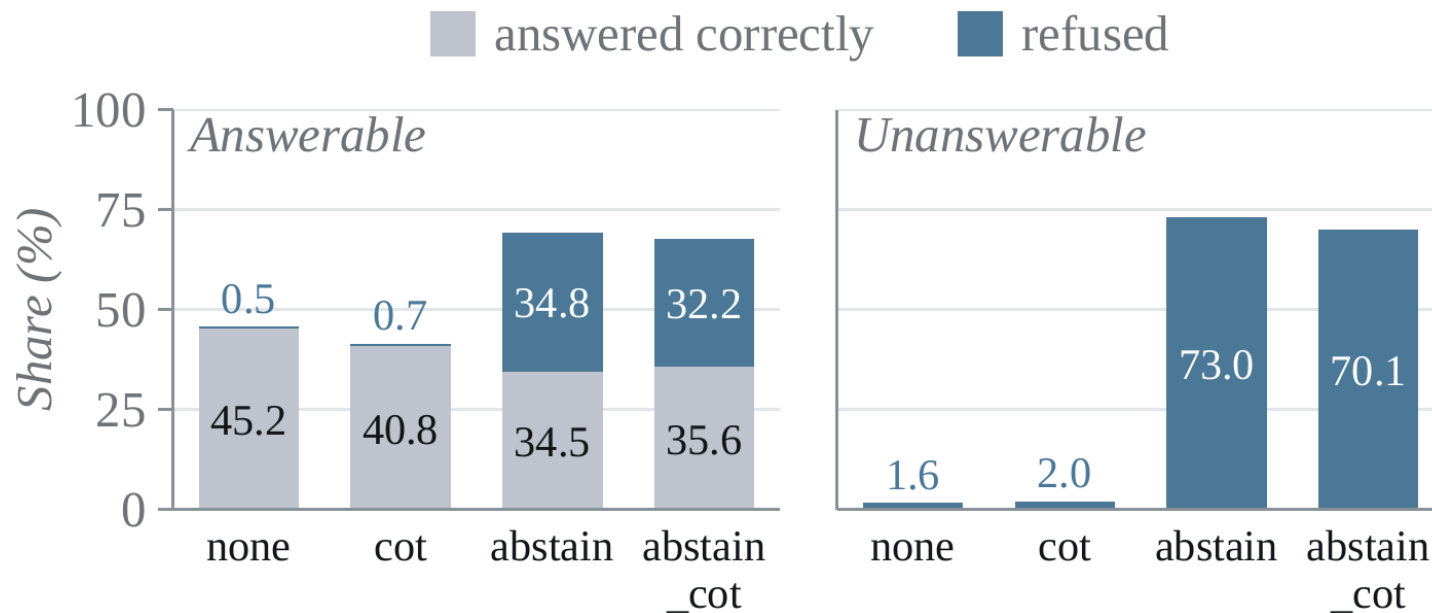


SELECTION

Gemma3-12B shows the same decline

- Single-hop falls from **60.3%** to **40.7%**, multi-hop from **35.4%** to **23.6%**.
- Added context still both repairs and breaks answers.

Calibration on Another Backbone



REASONING

Prompt effects depend on the model

- Adding CoT to abstain gives Gemma3-12B **1.1 points** of answerable accuracy, against 9.2 for Qwen3-VL-8B.
- Correct refusal falls 2.9 points for Gemma and rises 9.4 for Qwen.

Representation Cost

	Acc.	Tokens per page			Latency (s)	
		Text	Vis.	Total	Prefill	Decode
T	29.0	499	0	499	1.4	5.7
TL	35.7	874	0	874	1.8	6.6
TV	49.6	500	459	959	—	—
TLV	49.5	874	459	1333	26.3	6.7
V	42.6	20	459	479	29.0	7.1

DEPLOYMENT

Latency follows visual processing, not token count

- Prefill is **29.0 s** per question with images alone, against 1.4 to 1.8 s for text.
- Embedded text plus images matches parser text plus images: **50.1% against 49.5%**, with no parser.

Reasoner Choice Under Budget

Reasoner	T	TL	TLV	V	GB
<i>Scale (Qwen3-VL, bf16)</i>					
2B	24.7	31.9	38.3	31.9	4.3
4B	32.3	38.6	50.4	42.4	8.9
8B	31.9	38.4	52.4	45.5	17.5
32B	35.5	43.9	59.6	56.7	66.7
<i>Quantized (Qwen3-VL)</i>					
8B, 8-bit	32.5	37.9	53.6	45.7	10.3~
8B, 4-bit	31.8	39.4	51.8	45.5	6.8~
32B, 4-bit	37.0	42.1	59.1	52.3	20.0~
<i>Other families (bf16)</i>					
InternVL3-8B	19.3	25.3	32.6	24.1	15.9
MiniCPM-V-4.5 (8B)	30.7	38.1	60.1	53.3	17.4
GLM-4.6V-Flash (9B)	25.9	32.9	42.7	30.9	20.6
Llama-3.2-11B-Vision	26.5	34.3	33.3	25.3	21.3
Gemma-3-12B	30.5	36.9	48.1	41.6	24.4

DEPLOYMENT

Family and precision matter as well as size

- Qwen3-VL-8B keeps **51.8%** at 4-bit, in about 6.8 GB of weights.
- The 4-bit 32B model reaches **59.1%** in about 20 GB.

From Findings to Design

The survey and the experiments both point at design principles. I put them together in a **deliberately simple system**, to see whether they hold up at the level of a whole pipeline rather than one controlled condition.

REPRESENTATION	Keep text and images together ▶ Every page read at the full TLV representation, with a strong parser and high-resolution rendering, since the two channels cover for each other's failures.
SELECTION	Gather broadly, present selectively ▶ Missing evidence is unrecoverable and moderate distraction is not, so read widely in retrieval-ranked order, but carry most of it forward as compact records rather than as full pages.
REASONING	Check whether the answer is supported ▶ Because the reasoner cannot judge sufficiency on its own, a separate step decides whether to accept the answer or keep reading.

No planner, no tool set, no query reformulation. **The point is to validate the principles, not to introduce a complex architecture.**

Simple Method Cost

	Turns	Tokens			Latency (s)		Peak
		Input text	Input visual	Output	Prefill	Decode	VRAM (GiB)
reader	11.5	5,030	816	697	1.8	46.8	7.2
synthesizer	1.2	9,307	2,018	463	3.8	34.7	8.1
arbiter	1.2	6,806	1,734	464	2.9	33.3	7.9
Per question, mean	13.8	76,669	13,767	9,092	28.5	617.5	8.2
Per question, median	9.0	43,113	9,830	6,078	15.8	385.0	8.0

- **13.8 turns** per question on average, 11.5 of them the reader.
- The cost is time, not memory: **10.8 minutes** mean, 6.7 median, with peak memory at 8.2 GiB.